Phase 5: Modernization intelligence

Phase 5 turns a captured architecture version into a reviewable modernization proposal. It does not edit source code or execute a modernization wave.

Start

1. Open ArchIntel at `http://127.0.0.1:8000` and sign in.
2. Complete a static analysis. Optionally collect real runtime sessions against that exact baseline.
3. In **Evidence & digital twin**, preview and save an architecture version. Review domain membership there if you have validated the candidate boundaries.
4. Open **Modernization intelligence**. Select the application and captured architecture version.
5. Choose a target strategy and describe the planning objective.
6. Click **Preview modernization plan**. Review the proposed decisions and validation result, then **Save modernization plan**.

Saving the same version, objective, strategy and rules returns the existing proposal. Changing the objective or strategy creates another immutable proposal. Preview does not save a proposal.

Understand the results

Decisions shows an action and blockers for each domain. **Explain why** traces the action to findings and supporting facts. Fact links open canonical provenance, including retained or expired span references. The export contains all evidence IDs; the UI shows at most 50 fact links per explanation.

Readiness & confidence separates architecture suitability from strength of evidence:

- Structural readiness preserves the existing baseline score unless domain membership was reviewed; reviewed domains are recomputed.
- Runtime readiness is unavailable when no mapped outgoing calls support a score.
- Combined readiness uses 60% structural and 40% runtime when runtime is available; otherwise it retains structural readiness.
- Observation gaps lower confidence rather than multiplying readiness by coverage.
- Confidence is an explainable heuristic index, not a probability of modernization success. Behavioral validation and data ownership remain unknown in Phase 5.

A high readiness score does not approve extraction. Sparse observations do not prove unused classes or independent services.

Targets and waves

Supported planning strategies are modular monolith, microservices, hybrid, cloud refactor, Java upgrade and framework upgrade. Modular monolith proposes in-process boundaries. Distributed targets leave deployment and contracts undecided pending review. Cloud and upgrade strategies require a compatibility inventory and receive manual-review decisions; this phase does not claim compatibility analysis.

Waves put dependency domains before their consumers. Cyclic groups remain together for remediation. Each wave includes readiness, confidence, difficulty, dependencies, prerequisites, blockers and a proposed impact. Impact remains a hypothesis until later source changes are validated.

Review

Open **Review**, choose **APPROVED_PLANNING**, **CHANGES_REQUESTED** or **REJECTED**, and provide a note. Review history records the actor, time and revision. If another review was saved, refresh before submitting again.

Planning approval records agreement with the proposal. It does not authorize code changes, extraction or deployment. To change the strategy or objective, create another proposal. To change boundaries or evidence, capture a new architecture version and generate a new proposal.

The latest captured version for the application defines current planning context. Older proposals show **SUPERSEDED** and cannot receive new reviews. They remain readable and exportable. The exported proposal is frozen; current context and review history are separate from it.

Graph and recovery

Graph shows a separate persistent planning graph linking assessments, decisions, findings, fact/evidence references, prerequisites, blockers and waves. It preserves the Phase 4 evidence graph. Select a node or click a relationship to explore a bounded neighborhood.

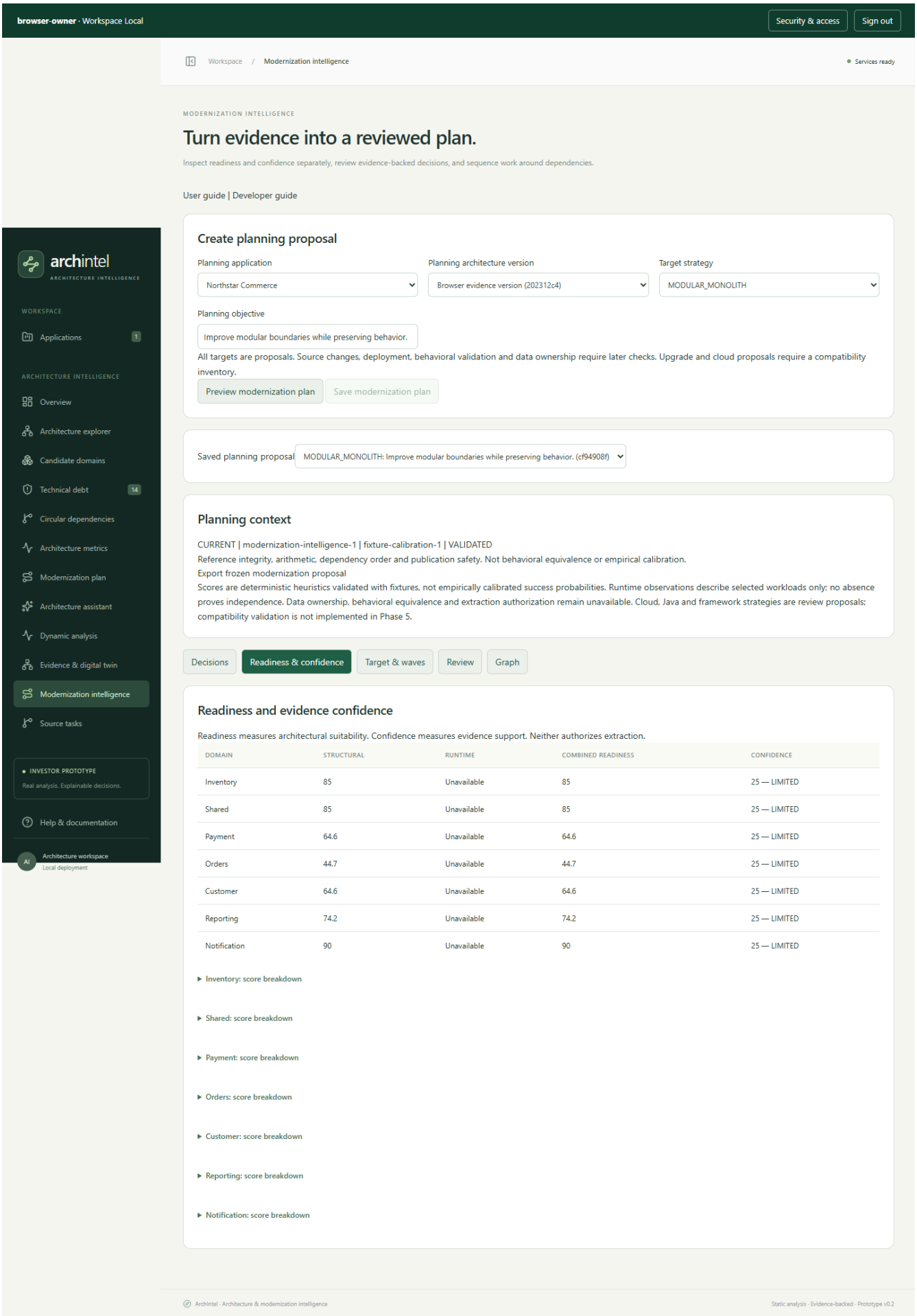
If Neo4j is unavailable, the saved proposal, explanations, reviews and exports remain available from PostgreSQL. Restore Neo4j and click **Rebuild planning graph**. The displayed projection status is the last attempt; opening the graph checks current content. Planning projections are retried manually in this release.

Permissions and limits

- Reading plans requires `ANALYSIS_READ`; creation and review require `ANALYSIS_WRITE`.
- Versions containing runtime evidence additionally require `RUNTIME_READ`, including explanations and exports.
- No model is invoked by this workflow. The existing AI gateway and user policy remain unchanged.
- Planning supports up to 500 domains and 30,000 dependency facts per captured version, with bounded graph display.
- Missing source/build linkage, behavioral contracts, data ownership and human change approval remain explicit prerequisites. No **EXTRACT_NOW** decision is published by Phase 5.

Suggested verification

Create a static-only plan: runtime readiness should be **Unavailable**, and observation gaps should be visible. Inspect an explanation and open its fact provenance. Approve the planning proposal, then capture a successor version: the old proposal should become historical and reject further reviews. Confirm the original static report and evidence export remain unchanged.


browser-owner · Workspace Local
Security & access
Sign out
Workspace / Modernization intelligence
Services ready
archintel
ARCHITECTURE INTELLIGENCE
WORKSPACE
Applications 1
ARCHITECTURE INTELLIGENCE
Overview
Architecture explorer
Candidate domains
Technical debt 14
Circular dependencies
Architecture metrics
Modernization plan
Architecture assistant
Dynamic analysis
Evidence & digital twin
Modernization intelligence
Source tasks
INVESTOR PROTOTYPE
Real analysis. Explainable decisions.
Help & documentation
Architecture workspace
Local deployment
MODERNIZATION INTELLIGENCE
Turn evidence into a reviewed plan.
Inspect readiness and confidence separately, review evidence-backed decisions, and sequence work around dependencies.
User guide | Developer guide
Create planning proposal
Planning application
Northstar Commerce
Planning architecture version
Browser evidence version (202312c4)
Target strategy
MODULAR_MONOLITH
Planning objective
Improve modular boundaries while preserving behavior.
All targets are proposals. Source changes, deployment, behavioral validation and data ownership require later checks. Upgrade and cloud proposals require a compatibility inventory.
Preview modernization plan
Save modernization plan
Saved planning proposal
MODULAR_MONOLITH: Improve modular boundaries while preserving behavior. (cf94908f)
Planning context
CURRENT | modernization-intelligence-1 | fixture-calibration-1 | VALIDATED
Reference integrity, arithmetic, dependency order and publication safety. Not behavioral equivalence or empirical calibration.
Export frozen modernization proposal
Scores are deterministic heuristics validated with fixtures, not empirically calibrated success probabilities. Runtime observations describe selected workloads only; no absence proves independence. Data ownership, behavioral equivalence and extraction authorization remain unavailable. Cloud, Java and framework strategies are review proposals; compatibility validation is not implemented in Phase 5.
Decisions
Readiness & confidence
Target & waves
Review
Graph
Readiness and evidence confidence
Readiness measures architectural suitability. Confidence measures evidence support. Neither authorizes extraction.
DOMAIN
STRUCTURAL
RUNTIME
COMBINED READINESS
CONFIDENCE
Inventory 85 Unavailable 85 25 — LIMITED
Shared 85 Unavailable 85 25 — LIMITED
Payment 64.6 Unavailable 64.6 25 — LIMITED
Orders 44.7 Unavailable 44.7 25 — LIMITED
Customer 64.6 Unavailable 64.6 25 — LIMITED
Reporting 74.2 Unavailable 74.2 25 — LIMITED
Notification 90 Unavailable 90 25 — LIMITED
▸ Inventory: score breakdown
▸ Shared: score breakdown
▸ Payment: score breakdown
▸ Orders: score breakdown
▸ Customer: score breakdown
▸ Reporting: score breakdown
▸ Notification: score breakdown
ArchIntel · Architecture & modernization intelligence
Static analysis · Evidence-backed · Prototype v0.2